

Workshop 4

1. Are either, both, or neither of the following two models GLMs? Explain.

$$\begin{aligned}\log(y_i) &= \beta_0 + \beta_1 x_i, & y_i &\sim \text{gamma}, \\ \log(y_i) &= \beta_0 + \beta_1 x_i, & \log(y_i) &\sim \text{gamma}.\end{aligned}$$

2. Suppose y_i represents the number of bacterial cells in a sample from a Petri dish and x_i is the amount of nutrient in the dish. Suppose further that:

$$\mathbb{E}(y_i) = a^2 + 2bax_i + b^2x_i^2, \quad i = 1, \dots, n.$$

Write down a suitable GLM for the data $\{(y_i, x_i)\}_{i=1}^n$.

3. The following table classifies a random sample of women and men according to their belief in an afterlife:

	Believer	Non-Believer
Female	435	147
Male	375	134

The data come from the US General Social Survey (1991), and the ‘non-believer’ category includes ‘undecided’.

Are there differences between males and females in their belief in an afterlife?

A null model for the data would be that belief does not depend on gender. Let i index the four cells of the table. If y_i denotes the observed count in cell (k, j) , where k indexes gender and j indexes belief status, then the expected count may be written

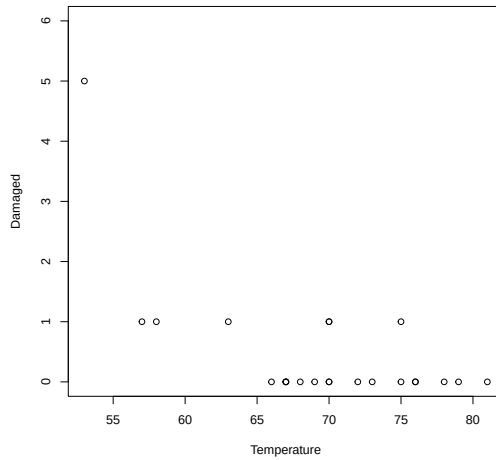
$$\mu_i \equiv \mathbb{E}(y_i) = n\gamma_k\alpha_j.$$

Here n is the total number of individuals sampled, γ_1 and γ_2 are the proportions of females and males in the population sampled, and α_1 and α_2 are the proportion of believers and non-believers in the population sampled, respectively.

For modelling purposes, it turns out that an appropriate response distribution to use here is the Poisson distribution. Show that this model can be written as a GLM with log link. Write out the full design matrix for the model and then a version in which the parameters are identifiable.

4. In January 1986, the space shuttle *Challenger* exploded shortly after takeoff. A subsequent investigation focused on the possibility that O-rings in the fuel tanks had failed due to the unusually low launch temperature (31°F), allowing a fuel leak. It turned out that data on O-ring failures from previous launches were available for a range of launch temperatures.

The plot below shows the number of failed O-rings (out of six) for launches at different temperatures. The first few rows of the dataset are also shown.



##	temperature	damaged	undamaged
## 1	53	5	1
## 2	57	1	5
## 3	58	1	5
## 4	63	1	5
## 5	66	0	6
## 6	67	0	6

Write down a suitable GLM for these data. Be sure to specify the link function, response distribution, and the form of the model matrix.

5. Show that the exponential distribution with probability density function given by

$$f(y; \lambda) = \lambda e^{-\lambda y}, \quad y > 0, \quad \lambda > 0,$$

belongs to the exponential family of distributions. Using the properties of the exponential family, show that

$$\mathbb{E}(Y) = 1/\lambda, \quad \text{var}(Y) = 1/\lambda^2.$$

6. A Bernoulli random variable Y takes the value 1 with probability p and 0 with probability $1 - p$. Its probability mass function is

$$f(y; p) = p^y (1 - p)^{1-y}, \quad y \in \{0, 1\}, \quad 0 < p < 1.$$

(a) Find $\mu = E(Y)$.

(b) Show that the Bernoulli distribution can be written in the general exponential family form.

7. Assume that $Z \sim \text{Bin}(n, \pi)$, i.e., Z is binomially distributed with n trials and probability π of success in each trial. Let $Y = Z/n$ denote the proportion (rather than the number) of successes. Note that $E(Y) = \pi$, which does not depend on n . The probability mass function of Y can be written as

$$f(y; \pi, n) = \binom{n}{ny} \pi^{ny} (1 - \pi)^{n - ny}, \quad y = 0, \frac{1}{n}, \frac{2}{n}, \dots, 1.$$

Show that the distribution of Y belongs to the exponential family of distributions and determine θ , $b(\theta)$, $a(\phi)$, and $c(y, \phi)$.

8. (a) Show that the Inverse Gaussian distribution with density given by

$$f(y; \mu, \gamma) = \sqrt{\frac{\gamma}{2\pi y^3}} \exp \left\{ -\frac{\gamma(y - \mu)^2}{2\mu^2 y} \right\}, \quad y > 0, \mu > 0, \gamma > 0,$$

where μ is the mean and γ is a scaling parameter, can be written in the form

$$f(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right\},$$

and determine θ , $b(\theta)$, $a(\phi)$, and $c(y, \phi)$.

- (b) Show that $\mathbb{E}(Y) = \mu$ and $\text{var}(Y) = \frac{\mu^3}{\gamma}$.

- (c) What is the canonical link function for this distribution?

9. Show that the t distribution, which is suitable for modelling heavy tail data and whose probability density function is given by

$$f(y; \nu) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\Gamma(\frac{\nu}{2})} \left(1 + \frac{y^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad y \in \mathbb{R}, \quad \nu > 0,$$

is not in the exponential family.

10. In a study of the presence of tumors in animals, suppose

$$Y_i \stackrel{\text{ind}}{\sim} \text{Poisson}(\mu_i), \quad \log(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad i = 1, \dots, n.$$

However, the observed response, Z_i , merely indicates whether each Y_i is positive, $Z_i = I(Y_i > 0)$. Define $\pi_i = \Pr(Z_i = 1)$. Then, the Z_i s follow a binomial GLM

$$Z_i \stackrel{\text{ind}}{\sim} \text{Bin}(1, \pi_i), \quad i = 1, \dots, n.$$

Show that the link function for this binomial model is given by

$$\log(-\log(1 - \pi_i)) = \mathbf{x}_i^\top \boldsymbol{\beta}.$$

11. Let y_i be independent random variables each with mean μ_i and variance $V(\mu_i)\phi$, where $\mu_i = g^{-1}(\mathbf{x}_i^\top \boldsymbol{\beta})$, g is a differentiable and strictly monotonic function, $\boldsymbol{\beta}$ a parameter vector, and $\mathbf{x}_i^\top$ is the i th row of the design matrix $\mathbf{X}$. Define

$$z_i = g'(\mu_i)(y_i - \mu_i) + \mathbf{x}_i^\top \boldsymbol{\beta}, \quad i = 1, \dots, n.$$

Find the expected value and covariance matrix of the vector $\mathbf{z} = (z_1, \dots, z_n)^\top$.

12. This question is about getting a feel for the IRLS method used to fit GLMs.

- (a) Find an expression for the value of β minimizing

$$\sum_{i=1}^n w_i (z_i - \beta x_i)^2$$

in terms of w_i , z_i , and x_i .

- (b) Consider fitting the following GLM

$$\mu_i \equiv \mathbb{E}(y_i) = \beta x_i, \quad y_i \sim \text{Poi}(\mu_i),$$

to the data:

x	1	2	3	4	5
y	0	1	3	7	4

Apply two steps of the IRLS scheme, by hand, in order to obtain an MLE of β . You will need to modify the default starting values for the IRLS, by changing the initial value used for $\hat{\mu}_1$ (why?): I suggest starting with $\hat{\mu}_1 = 1$.

- (c) Obtain an estimate of the standard error of $\hat{\beta}$.